QoS with Proposal for RISC-V in Cloud

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Agenda

- Requirements in Cloud
- RISC-V QoS
- ARM MPAM Architecture
- MPAM Monitoring
- MPAM Partitioning
- RISC-V Proposal

Requirements in Cloud

Requirements of Mixed Deployment in Cloud

Mixed-workload environments have a critical need for resource control and monitoring. including but not limited to:

- Special latency-sensitive workload require appropriate resources
- Customized strategies are needed to dynamically adjust resources
- Set the correct request and limit according to the resource requirements of the workload
- Real-time detection of noisy neighbors based on monitoring data

RISC-V QoS

RISC-V QoS Extensions and Ssqosid

RISC-V's QoS system includes two extensions:

- RISC-V Quality-of-Service (QoS) Identifiers (Ssqosid)
- RISC-V Capacity and Bandwidth Controller QoS Register Interface (CBQRI)

The Ssqosid extension adds the srmcfg register in the CSR, which can assign Resource Control ID (RCID) and Monitoring Counter ID (MCID) to each CPU Core.

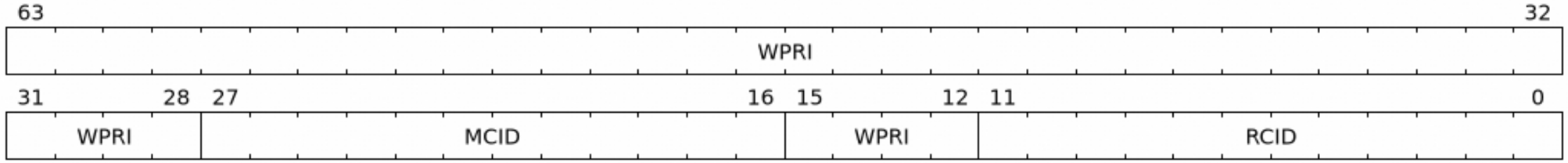


Figure 1. Supervisor Resource Management Configuration (`srmcfg`) register for SXLEN=64

Each request made to a shared resource carries the RCID and MCID:

- The resource controller uses the RCID in the request to determine the resource allocation to perform (e.g., cache occupancy limits, memory bandwidth limits, etc.).
- The resource controller uses the MCID in the request to identify the ID of the counter to monitor resource usage (e.g., cache occupancy, memory bandwidth, etc.).

RISC-V QoS CBQRI

The main purpose of CBQRI extension：

- Capacity usage monitoring
- Capacity allocation control
- Bandwidth usage monitoring
- Bandwidth allocate control

The RISC-V CBQRI extension includes the following two registers for Capacity and Bandwidth

- Capacity-controller QoS Registers interface
- Bandwidth-controller QoS Registers interface

Offset	Name	Size	Description	Optional?
0	cc_capabilities	8	Capabilities	No
8	cc_mon_ctl	8	Usage monitoring control	Yes
16	cc_mon_ctr_val	8	Monitoring counter value	Yes
24	cc_alloc_ctl	8	Capacity allocation control	Yes
32	cc_block_mask	BMW/8	Capacity block mask	Yes
N	cc_cunits	8	Capacity units count	Yes

Offset	Name	Size	Description	Optional?
0	bc_capabilities	8	Capabilities	No
8	bc_mon_ctl	8	Usage monitoring control	Yes
16	bc_mon_ctr_val	8	Monitoring counter value	Yes
24	bc_alloc_ctl	8	Bandwidth allocation control	Yes
32	bc_bw_alloc	8	Bandwidth allocation	Yes

- Cache and memory Monitoring control reg: set the request type and MCID;
- Capacity alloction control reg: set the RCID and resource uper limit(value setted in cc_cuints)
- Bandwitdh allocation control reg: ret RCID and reserve bandwith resource for workload;
- Bandwidth allocation reg: Set bandwidth ratio and remaining bandwidth resource weight（1/256）

ARM MPAM Architecture

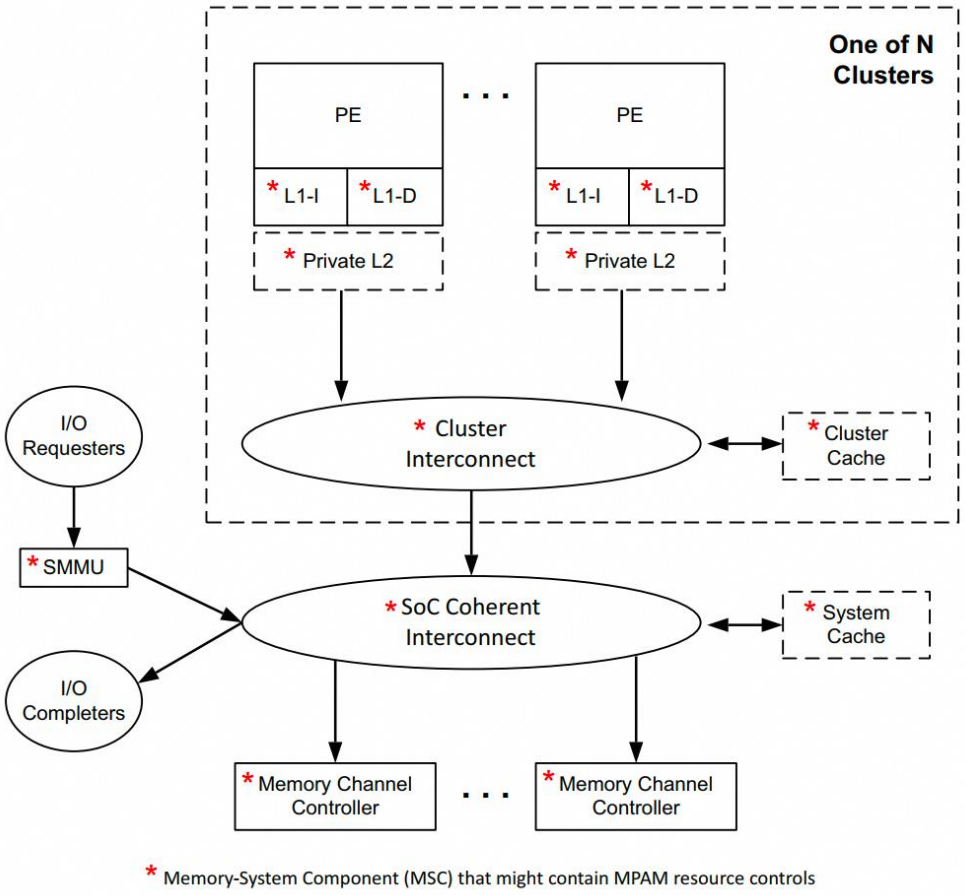
ARM MPAM Architecture

Each resource component contains a MSC that manages resource monitoring and allocation

- L1I and L1D cache resource
- Private L2 cache
- Cluster interconnect -- cluster cache
- Soc coherent interconnect – system cache
- SMMU for IO device
- Memory controller

CPU agents and non-CPU agents use the following tags for resource monitoring and allocation

- PARTID: are used for control of shared platform resources, such as L3 cache occupancy or memory bandwidth
- PMG: are used for monitoring of shared platform resource utilization



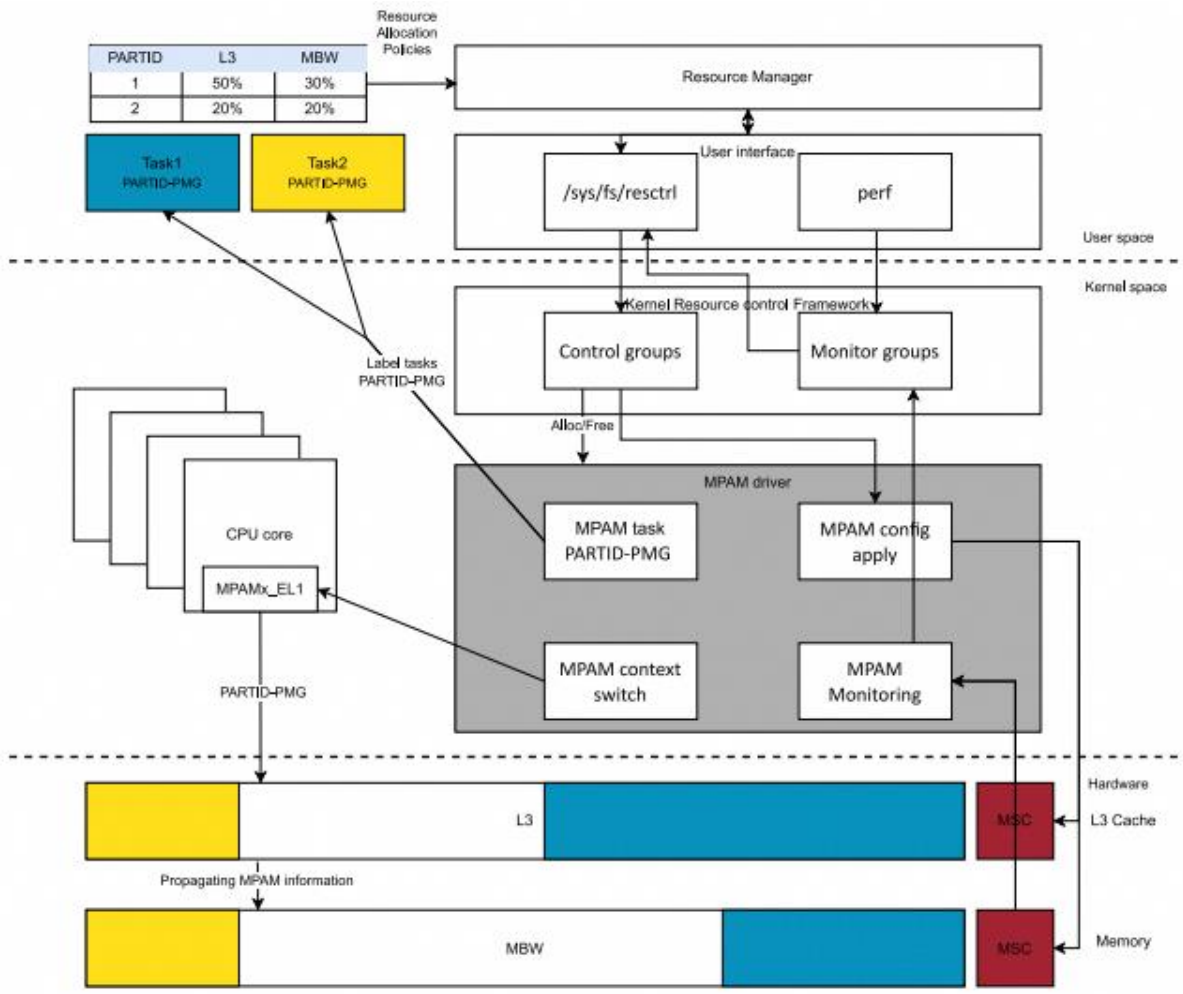
- Resource monitoring
- Resource partitioning

→ Cache and memory bandwidth

The Overall Framework of MPAM

- The MPAM system has three main layers:
  - The first layer is the user-mode resctrl interface layer； User can create control and monitor gorups by resctrl sysfs interfaces
  - The second layer is the kernel-mode general abstraction layer, which is responsible for compatibility with hardware of different architectures.
  - The last layer is the driver layer. The mpam driver is responsible for setting configuration to the hardware and reading monitoring data

- For RISC-V, the second layer support is currently being pushed, and other support has not yet been seen.

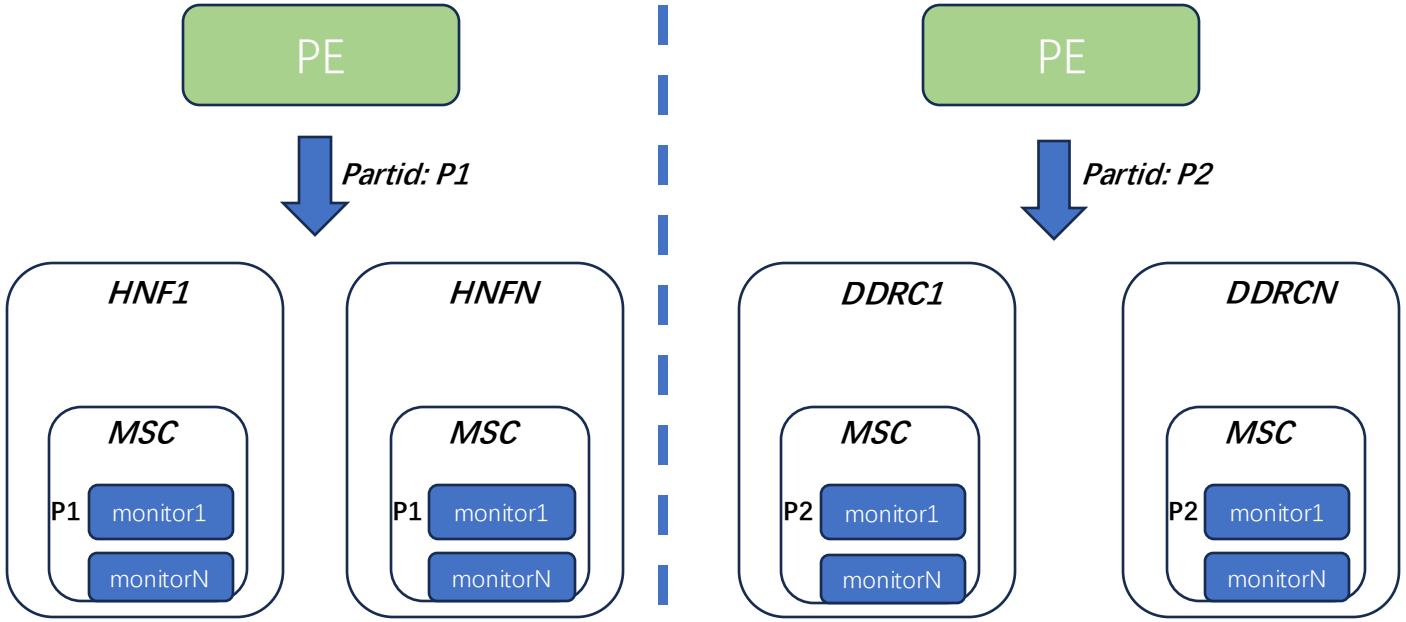


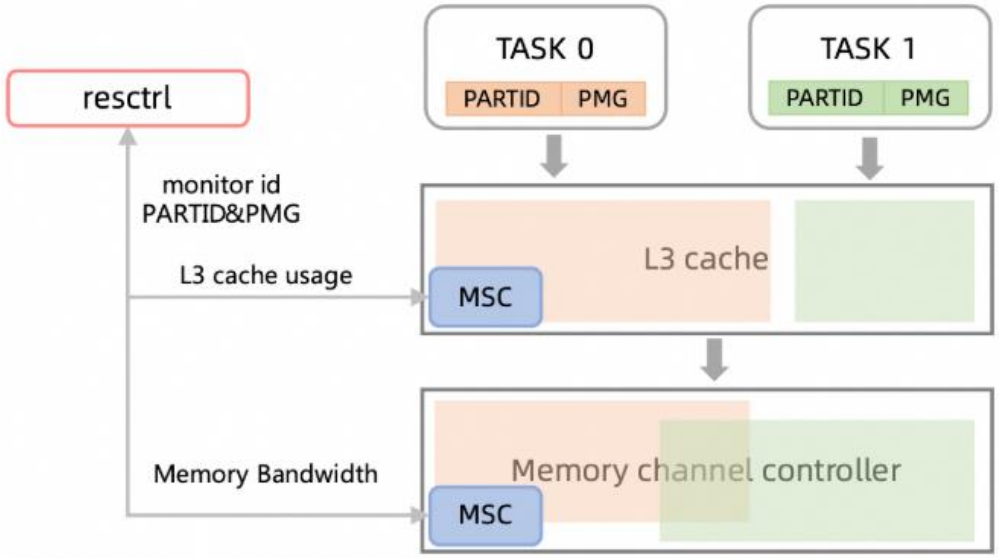
MPAM Monitoring

MPAM Monitoring

Resource monitoring is mainly divided into two categories

- Cache storage usage monitors
  - The request sent by the Core carries the partid, and the request is forwarded to the HNF.
  - The monitor in the HNF is responsible for monitoring the cache utilization.
- Memory-bandwidth usage monitors
  - Memory bandwidth monitoring is similar to cache monitoring





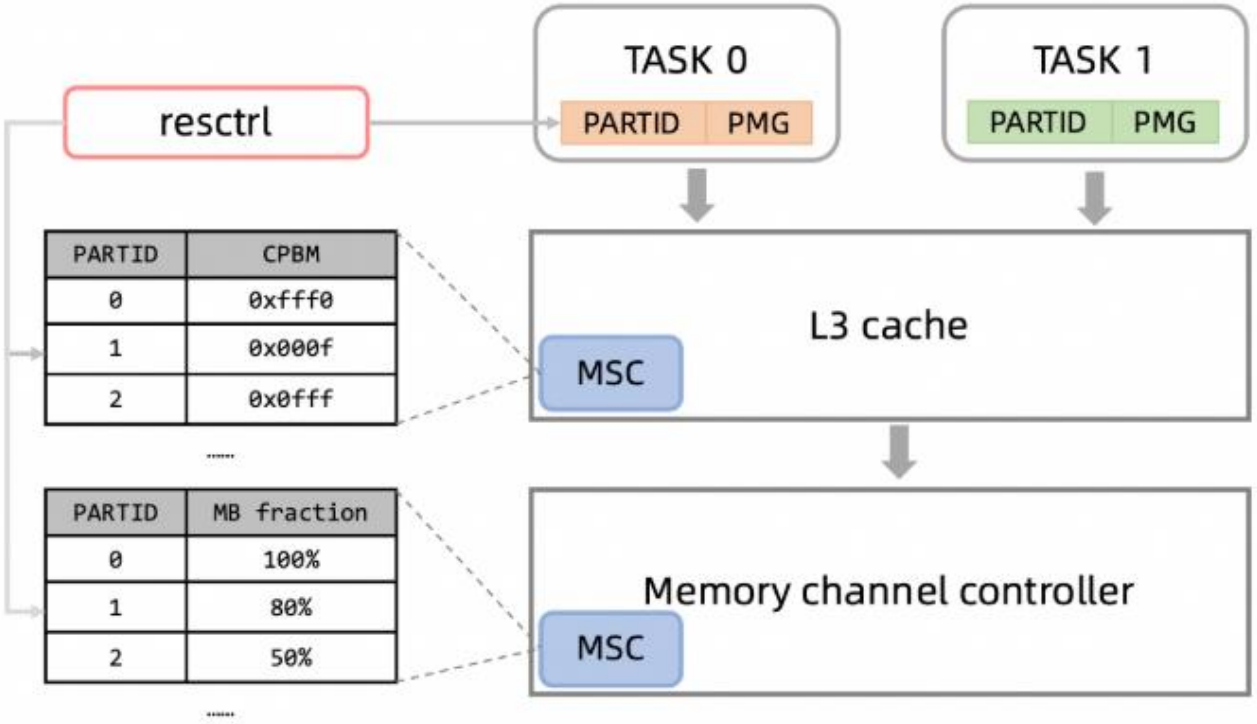
MPAM Partitioning

MPAM Partitioning

Resource partitioning is mainly divided into two categories

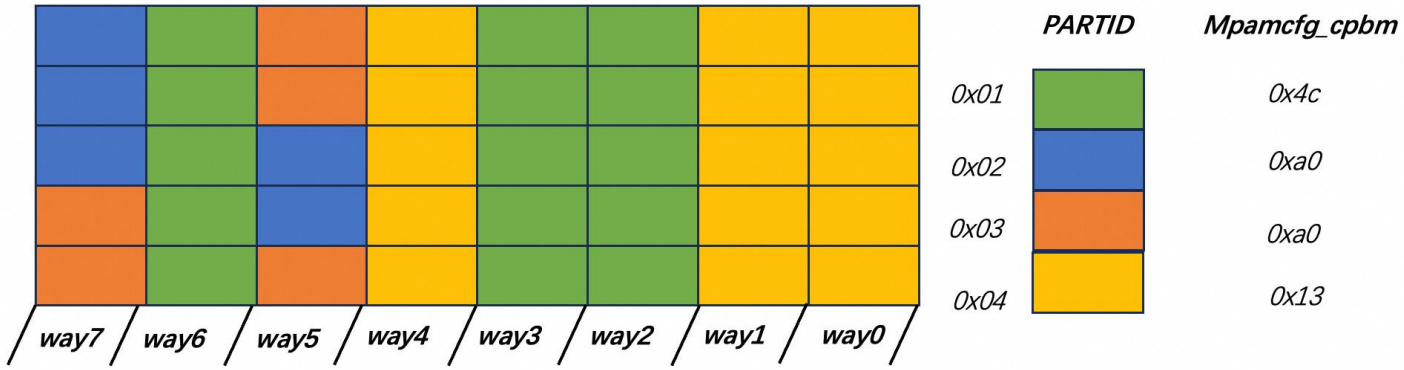
- Memory bandwidth partitioning
  - Memory-bandwidth portion partitioning.
  - Memory-bandwidth minimum and maximum partitioning.
  - Memory-bandwidth proportional-stride partitioning.
  - Priority partitioning

- Cache partitioning
  - Cache portion partition
  - Cache max-capacity partition



Cache Portion Partitioning

- Now support SLC isolation by way in our real environment
- The Cache way that a certain PARTID can be allocated to can be controlled.
- For example, PARTID 1 in the figure can be allocated to way2, way3, and way6.
- You can allocate cache portions for exclusive use of a partition or shared between two or more partitions

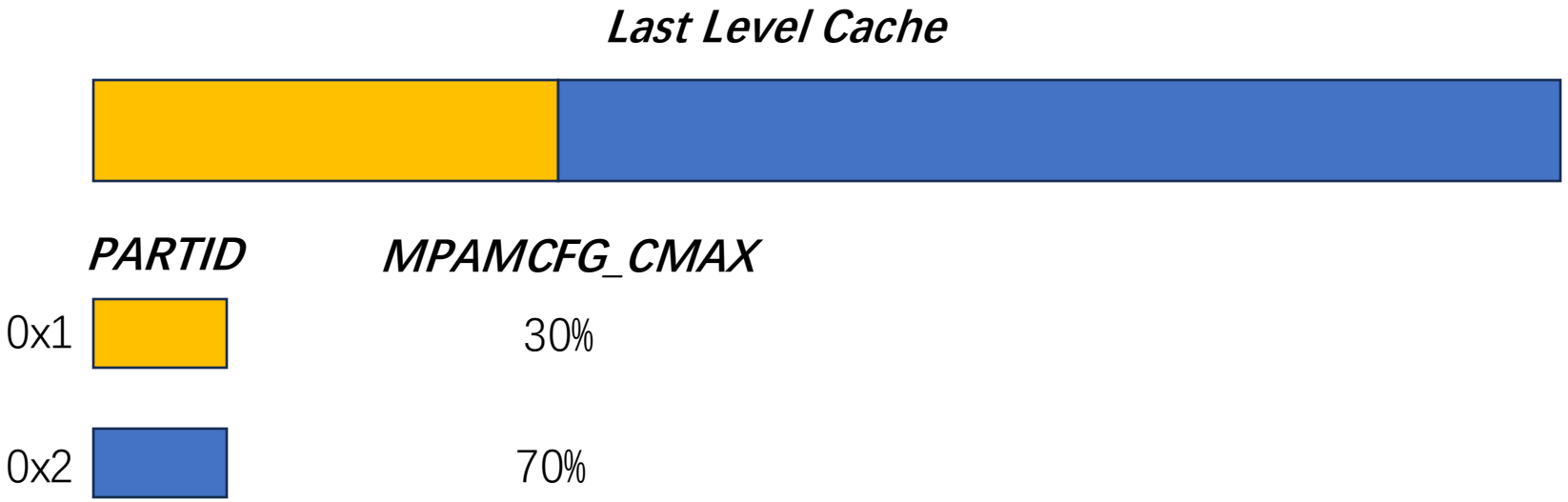


partid 1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	Exclusive bitmask
partid 2	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	

partid 1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	Shared bitmask
partid 2	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	

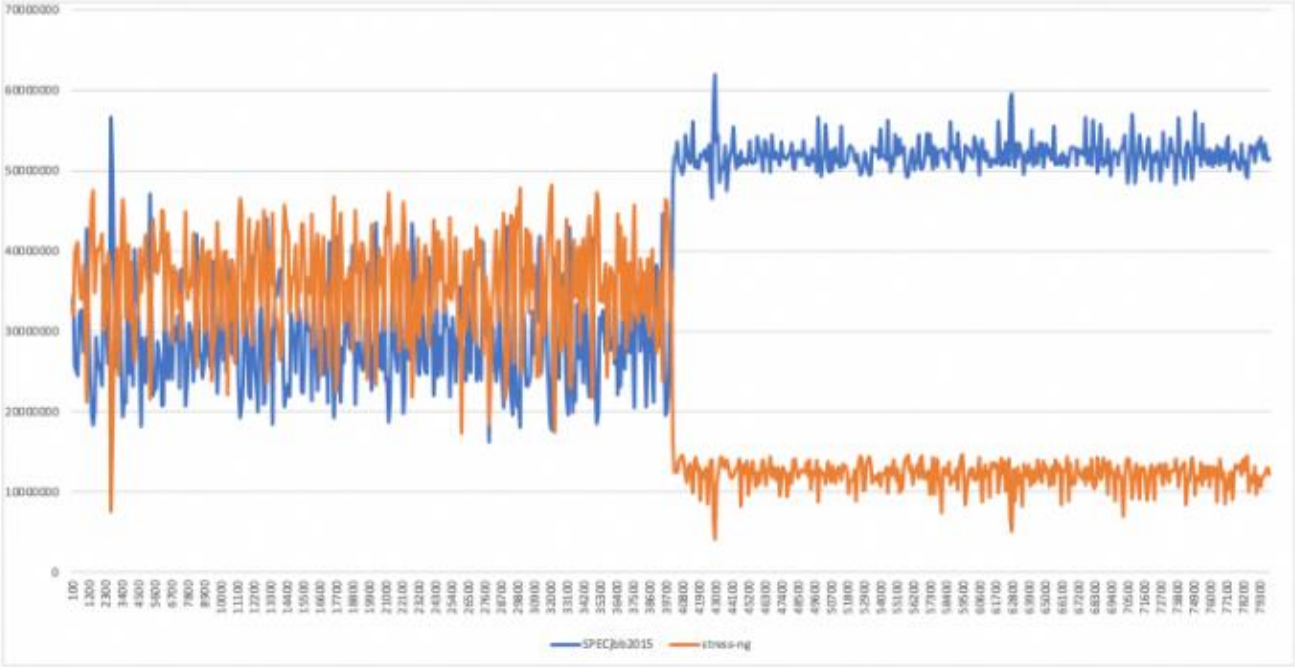
Cache Maximum-Capacity Partitioning

- Support SLC to be controlled according to the maximum capacity
- Each PARTID can specify its own maximum capacity used through MPAMCFG_CMAX
- We can increase or decrease the capacity at a granularity of 1/128
- When allocating a new cache line, it will be judged whether the capacity exceeds the limit of CMAX, if yes, a cache line owned by this partid will be replaced.



Use Case of Cache Partitioning In Cloud

- Mixed deployment is an important application scenario for MPAM resource isolation.
- The SPECjbb 2015 and stress-ng programs are used to simulate the Java-based online and offline jobs widely used in cloud environments.
- LLC partitioning isolates online and offline workloads to mitigate resource contention and ensure stabl e performance for the online business.



Memory Bandwidth Partitioning

MPAM supports memory bandwidth partitioning through both proportional shares and maximum/minimum limits.

- Memory-bandwidth portion partitioning.
- Memory-bandwidth minimum and maximum partitioning.

```
[test@host /sys/fs/resctrl/sigma-job]
$ cat schemata
MB:0=0000080;1=0000080;2=0000080;3=0000080
L3:1=ffffffff;98=ffffffff;195=ffffffff;292=ffffffff
```

Memory-bandwidth minimum and maximum partitioning.

- Minimum: If a PARTID's bandwidth falls below its configured minimum level (Min watermark) within a time window, requests from that PARTID will be prioritized for scheduling to ensure its real-time bandwidth remains above the minimum level.
- Max Softlimit: The softlimit setting indicates that the maximum limit can be exceeded if there is available memory bandwidth.
- Max Hardlimit: The hardlimit setting enforces a strict maximum that cannot be exceeded.
- You can increase memory bandwidth configuration with 1/256 granularity

Priority Partitioning

priority-based requests scheduling

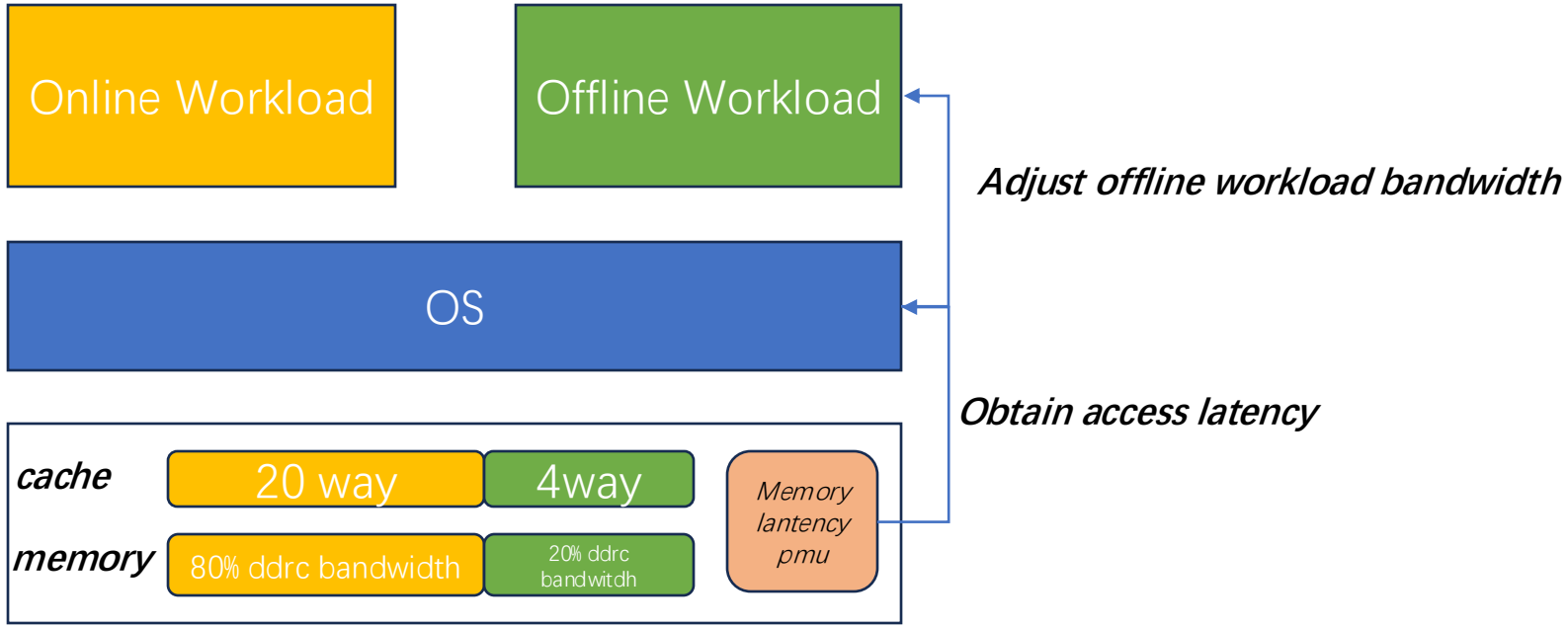
- A PARTID may be assigned priorities for each component in the memory system that implements a priority partitioning control.

- Normally, Priority Partitioning requires the support of the DDRC controller and consistent interconnection.
- The request has a higher priority and will receive priority scheduling support within the DDRC controller.
- At present, we do not have hardware that actually supports this feature and cannot use it in actual scenarios.

Use Case of Memory Bandwidth Partitioning

In e-commerce mixed-workload scenarios, MBA (Memory Bandwidth Allocation) and CAT (Cache Allocati on Technology) are used to isolate online and offline workloads.

- Use CAT to isolate cache resources by restricting offline tasks to 4 way cache
- Use MBA to isolate ddrc bandwidth by restricting offline tasks to 20% ddrc bandwidth
- We usually use memory bandwidth partition and memory access latency monitoring together
- If the online memory access lantency is too high, we will suppress the offline bandwidth with a certain granularity(1/256).



Comparison between CBQRI and MPAM

Partitioning features	MPAM support?	CBQRI support?
Cache portion partitioning	✅	✅
Cache minimum capacity partitioning	✅	✅
Cache maximum capacity partitioning	✅	✅
Memory bandwidth minimum partitioning	✅	✅
Memory bandwidth maximum partitioning	✅	✅
Memory bandwidth proportional stride partitioning	✅	✅
Priority partitioning	✅	❌
FLUSH_RCID	❌	✅

Comparison:

- MPAM supports priority partition, which CBQRI does not yet support
- CBQRI supports flush the cache line for a specific RCID, but MPAM not yet support
- If the maximum bandwidth of MPAM is not fully utilized, the remaining resources can be given to other processes, but CBQRI will directly reserve resources.

RISC-V Proposal

RISC-V Proposal

Proposal1:

Motivation:

- The current RISC-V Quality of (QoS) mechanisms generally lack explicit request prioritization or "priority partition" based on Requestor ID (RCID). This means all requests, regardless of their source or importance, are often treated with similar weight.

Solution:

- To integrate robust priority-aware QoS, we propose:

1. Request-level Priority Tagging:
    1. Introduce a dedicated 'priority' field for each request issued by an RCID.
    2. This field will explicitly convey the urgency/importance of the request as it travels through the system.
2. Configurable Per-RCID Priority Assignment:
    1. Extend RCID control registers to include a configurable priority bit-field.
    2. This allows system software (e.g., OS) to assign a default priority level for all requests originating from that specific RCID.

Q & A